

Game-Guided Tactical Corridor Learning for Autonomous Racing Overtakes

Shuyi Gao^a, Ren Jin^a, Defu Lin^a

^a*School of Aerospace Engineering, Beijing Institute of Technology, Beijing, China*

Abstract

Interactive overtaking requires more than choosing a passing side: the tactical decision must persist as the opponent responds and must expose enough space for a downstream planner to execute it. Direct game solvers capture the interaction but are costly online; discrete maneuver labels and single trajectories do not provide a persistent planner contract; and end-to-end policies hide whether a decision is executable. This paper introduces a game-guided tactical-corridor policy that makes a time-varying feasible set, rather than an actuator command, the learned decision variable. A Stackelberg game with iterative best responses supplies an action prior and a robust-value target to Truncated Quantile Critics, amortizing repeated interaction reasoning into one policy evaluation. The policy outputs a passing-side commitment, speed cap, and Frenet corridor; a hysteretic state machine and deterministic carver preserve the selected side and reject collapsed intervals. Candidate actions are projected through the identified tire-force feasibility model, and a common corridor-constrained path-speed planner and model predictive contouring controller produce throttle, brake, steering, and gear commands. The method was evaluated on a hardware-in-the-loop testbed in which the complete algorithm ran on the target Neousys RGS-8805GC computer and communicated through two Controller Area Network (CAN) channels with an EAV24 simulator on a separate server. The plant configuration includes subsystem-level powertrain, brake, steering, suspension, aerodynamic, thermal-tire, and sensor dynamics. Across 1000 closed-loop evaluation laps, the full method achieved a 94.7% overtaking success rate (95% exact binomial confidence interval, 93.1–96.0%) with no observed collision or track-boundary violation. Matched comparisons and component ablations attribute the gain primarily to learned corridor shaping, game-derived guidance, and persistent side commitment. The evidence is limited to two-vehicle engagements with simulator-provided state and opponent information.

Keywords: Autonomous racing, game-guided reinforcement learning, tactical corridor, interactive overtaking, hardware-in-the-loop simulation

1. Introduction

Autonomous racing exposes a control system to a regime in which tactical interaction and tire-force limits evolve on comparable time scales. A trailing vehicle must select an overtake before the available space closes, yet that decision is useful only if the downstream planner can construct a continuous path and the controller can realize it through the physical actuator interface. Full-scale events such as the Abu Dhabi Autonomous Racing League (A2RL) make this coupling practically consequential because racing software runs on vehicle hardware, observes other non-cooperative agents, and must return drive-by-wire commands under a hard real-time loop [3, 17]. The central control problem is therefore not merely collision avoidance or low-level tracking. It is how to turn opponent-aware tactical reasoning into a passing opportunity that remains geometrically explicit, temporally persistent, and executable under repeated replanning.

1.1. Related work

Existing racing systems represent tactical intent in three main ways. Rule-based state machines and maneuver libraries are transparent and have been deployed successfully,

but their fixed thresholds require extensive track-specific tuning [1, 2, 5]. Dynamic-game formulations couple the competitors' plans through best responses or equilibrium concepts, including Stackelberg and potential games [6, 7, 8, 9]. Their strategic semantics are attractive, although repeated equilibrium computation and sensitivity to opponent models complicate execution at racing rates. Reinforcement learning can instead amortize repeated interaction and has achieved strong racing performance [12, 13, 14, 15]. Recent Bayesian and hybrid formulations add model uncertainty or learned strategic guidance [10, 11]. End-to-end policies, however, generally hide the geometric feasibility assumptions needed by a model-based planner and controller.

Hierarchical approaches partially address this gap by separating tactical choice from trajectory generation [16, 17]. In many implementations, the interface remains a discrete mode, a target point, or a single predicted trajectory. These representations do not state how much track space is reserved, when a passing side becomes committed, or whether the remaining region stays useful under the next receding-horizon update. Treating the opponent only as a predicted moving obstacle can therefore produce a locally feasible but strategically frozen plan. Conversely, directly learning steering and longitudinal commands removes the opportunity for deterministic geometric and actua-

tor checks. These alternatives leave three linked gaps: repeated game solutions are too expensive for the tactical rate, reward-only learning has weak structural guidance for rare passing windows, and transient tactical outputs can switch sides or collapse before the planner finishes the maneuver. A time-varying Frenet corridor provides the missing interface. It exposes the chosen passing opportunity as explicit inequalities while leaving path regularity, dynamic feasibility, and actuator tracking to established optimization modules.

1.2. Contributions and paper organization

This paper develops a game-guided tactical corridor policy for interactive overtaking. A Stackelberg game with iterative best responses (IBR) provides the strategic decomposition, a prior action, and an auxiliary robust-value target. A Truncated Quantile Critics (TQC) policy amortizes this computation and selects continuous corridor parameters together with a persistent passing side. A deterministic state machine and corridor carver then convert the policy output into auditable bounds for a common execution stack. The downstream stack is deliberately held fixed in all comparisons. It serves as the execution layer needed to test the tactical contribution rather than as a separate claim of novelty.

The contributions are threefold.

1. We formulate tactical overtaking as interaction-aware feasible-set selection and derive a game-guided TQC objective that transfers both an IBR-consistent action prior and a robust game-value target. This construction amortizes repeated Stackelberg reasoning while limiting optimistic value estimates for rare, high-return passing sequences.
2. We introduce a planner-compatible hybrid interface that combines a learned time-varying corridor with hysteretic side commitment, deterministic opponent-footprint carving, and physical-feasibility projection. It prevents update-to-update side switching and corridor collapse while publishing lateral bounds and a speed cap that can be inspected independently of the learned policy.
3. We evaluate the tactical method in a target-hardware HIL loop. The complete stack runs on the A2RL RGS-8805GC computer and exchanges vehicle, opponent, race-control, and four-channel actuator data with an EAV24 plant through two Controller Area Network (CAN) channels. Matched baselines and ablations test tactical effectiveness, component necessity, and closed-loop failure modes.

Section 2 defines the decision variables and common feasibility models. Section 3 presents the game-guided corridor policy. Section 4 summarizes the common corridor-constrained execution layer. Section 5 reports the HIL evaluation and Section 6 discusses the evidence and its limitations.

2. Problem Formulation and Feasibility Models

Consider an ego vehicle and a non-cooperative opponent on a closed track. At each tactical update, the ego receives their

measured states, local track geometry, and race-control status. It must publish a mode, a committed passing side, a speed cap, and lateral bounds over a finite spatial horizon. The output is admissible only if the resulting corridor lies within the track, excludes the opponent footprint with the prescribed clearance, and contains a path-speed reference compatible with the identified tire-force envelope. This section defines the common vehicle, Frenet, and acceleration-limit models used to test those conditions. These reduced-order models belong to the algorithm; the HIL plant described in Section 5.1 is independently configured and is not integrated from these equations.

Let the continuous-time controller state and input be

$$\mathbf{x} = [X, Y, \psi, v_x, v_y, r, \delta, s_p]^\top, \quad \mathbf{u} = [a, \dot{\delta}]^\top, \quad (1)$$

where (X, Y) is the inertial position, ψ is yaw angle, (v_x, v_y) are body-frame longitudinal and lateral velocities, r is yaw rate, δ is front steering angle, and s_p is an auxiliary progress state used by the simulation framework for lap counting. The input a is the commanded longitudinal acceleration and $\dot{\delta}$ is the steering-rate command. The rigid-body kinematics are

$$\begin{aligned} \dot{X} &= v_x \cos \psi - v_y \sin \psi, & \dot{Y} &= v_x \sin \psi + v_y \cos \psi, \\ \dot{\psi} &= r, & \dot{s}_p &= v_x. \end{aligned} \quad (2)$$

The front and rear slip angles are

$$\alpha_f = \arctan\left(\frac{v_y + l_f r}{v_x + \varepsilon_v}\right) - \delta, \quad (3)$$

$$\alpha_r = \arctan\left(\frac{v_y - l_r r}{v_x + \varepsilon_v}\right), \quad (4)$$

where the small positive ε_v regularizes the low-speed denominator. With linear lateral tire stiffness,

$$F_{yf} = -C_f \alpha_f, \quad F_{yr} = -C_r \alpha_r. \quad (5)$$

The resulting Newton–Euler equations are

$$\dot{v}_x = a + r v_y - \frac{1}{m} F_{yf} \sin \delta, \quad (6)$$

$$\dot{v}_y = \frac{1}{m} (F_{yf} \cos \delta + F_{yr}) - r v_x, \quad (7)$$

$$\dot{r} = \frac{1}{I_z} (l_f F_{yf} \cos \delta - l_r F_{yr}). \quad (8)$$

The steering angle evolves as $\dot{\delta} = u_\delta$ where u_δ is the steering-rate input in (1). Equations (2)–(8) define

$$\dot{\mathbf{x}} = f_c(\mathbf{x}, \mathbf{u}), \quad \mathbf{x}_{k+1} = f_d(\mathbf{x}_k, \mathbf{u}_k), \quad (9)$$

where f_d is the fourth-order Runge–Kutta discretization with sampling time T_s . Numerical parameter values are collected in Table A.1 in Appendix A.

The linear lateral-force relation in Eq. (5) is the local small-slip approximation used by the reduced-order controller model. The load-dependent nonlinear characteristics used to construct the physical feasibility limits are represented by the pure-slip Magic Formula

$$\begin{aligned} \zeta_j &= B_j q_j - E_j (B_j q_j - \arctan(B_j q_j)), \\ F_j(q_j, F_z) &= D_j (F_z) \sin(C_j \arctan(\zeta_j)), \quad j \in \{x, y\}, \end{aligned} \quad (10)$$

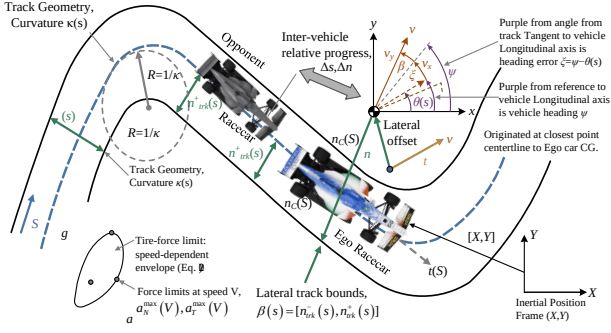


Figure 1: Vehicle and track model in the Frenet frame.

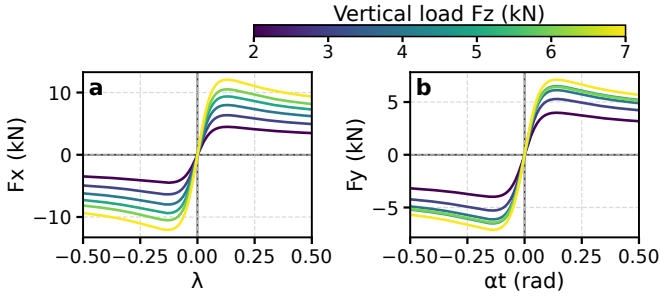


Figure 2: Load-dependent pure-slip tire-force characteristics. (a) Longitudinal force versus slip ratio. (b) Lateral force versus tire slip angle. All six curves in each panel retain the complete vertical-load sweep from 2 to 7 kN. Their nonlinear saturation supplies the physical feasibility limits used by the tactical projection and common executor; it is a supporting constraint, not the tactical decision variable.

where $q_x = \lambda$ is longitudinal slip ratio, $q_y = \alpha_t$ is tire slip angle, and $D_j(F_z)$ captures the load-dependent peak force. Figure 2 shows the corresponding force saturation and load sensitivity; the parameters are collected in Table A.1 in Appendix A.

Let $\mathbf{r}_c(s) \in \mathbb{R}^2$ be the track centerline parameterized by arc length $s \in [0, L_{\text{trk}})$. Its tangent angle is $\theta(s)$ and its curvature is $\kappa(s) = d\theta/ds$. Define the tangent and normal unit vectors as

$$\mathbf{t}(s) = [\cos \theta(s), \sin \theta(s)]^\top, \quad \mathbf{n}_c(s) = [-\sin \theta(s), \cos \theta(s)]^\top. \quad (11)$$

Any vehicle pose sufficiently close to the track is represented by

$$\mathbf{p} = \begin{bmatrix} X \\ Y \end{bmatrix} = \mathbf{r}_c(s) + n \mathbf{n}_c(s), \quad \xi = \psi - \theta(s), \quad (12)$$

where n is the signed lateral offset and ξ is the heading error. The body velocity projected onto the Frenet axes is

$$v_s = v_x \cos \xi - v_y \sin \xi, \quad v_n = v_x \sin \xi + v_y \cos \xi. \quad (13)$$

Differentiating (12) gives the exact kinematic relation

$$\dot{s} = \frac{v_s}{1 - \kappa(s)n}, \quad (14)$$

$$\dot{n} = v_n, \quad (15)$$

$$\dot{\xi} = \dot{\psi} - \kappa(s)\dot{s}. \quad (16)$$

The wrapped progress difference used in tactical reasoning is

always computed on this same coordinate,

$$\Delta s = ((s_o - s_e + L_{\text{trk}}/2) \bmod L_{\text{trk}}) - L_{\text{trk}}/2, \quad \Delta n = n_o - n_e. \quad (17)$$

For the path planner it is more convenient to use s as the independent variable and to describe a path by $\mathbf{z}(s) = [n(s), \xi(s)]^\top$. Let $\omega(s)$ denote the curvature-like steering variable of this geometric path. The spatial Frenet model is

$$\mathbf{z}'(s) = \mathbf{f}_s(\mathbf{z}(s), \omega(s); s) = \begin{bmatrix} (1 - \kappa n) \tan \xi \\ (1 - \kappa n) \omega / \cos \xi - \kappa \end{bmatrix}. \quad (18)$$

At planning iteration j , this model is linearized about the previous path iterate $(\bar{\mathbf{z}}_k, \bar{\omega}_k)$ at each station s_k ,

$$\mathbf{z}'(s_k) \approx \mathbf{A}_k \mathbf{z}_k + \mathbf{B}_k \omega_k + \mathbf{b}_k, \quad (19)$$

where $\mathbf{A}_k$, $\mathbf{B}_k$ and $\mathbf{b}_k$ are the Jacobian and affine residual of $\mathbf{f}_s$. This single linearized Frenet model is the one used by the second-order cone program (SOCP) in Sec. 4; no new lateral or heading variables are introduced later.

The nonlinear tire curves define pure-slip force capacities at each vertical load. After aggregation over the four wheels under the speed-dependent normal loads, load transfer, and aerodynamic downforce, they yield the tangential and normal acceleration limits $a_T^{\max}(V)$ and $a_N^{\max}(V)$ shared by the tactical feasibility check, speed planner, and model predictive contouring controller (MPCC). Combined tire-force use is approximated by the super-elliptic feasibility envelope

$$\left| \frac{a_N}{a_N^{\max}(V)} \right|^p + \left| \frac{a_T}{a_T^{\max}(V)} \right|^p \leq 1, \quad (20)$$

with exponent $p = 0.75$. Given a path curvature κ_p and speed V , the normal acceleration is $a_N = \kappa_p V^2$, and the remaining tangential acceleration is

$$a_T^{\text{avail}}(V, a_N) = a_T^{\max}(V) \left[1 - \left(\frac{|a_N|}{a_N^{\max}(V)} \right)^p \right]^{1/p}. \quad (21)$$

Aerodynamic drag is modeled as

$$a_{\text{drag}}(V) = \frac{1}{m}(c_{d,2}V^2 + c_{d,1}V + c_{d,0}), \quad (22)$$

and is added with the acceleration or braking sign convention stated in Sec. 4. The same envelope is used in Section 3 to project game and TQC actions before value evaluation and in Section 4 as a hard physical admissibility constraint for speed profiling and control.

3. Game-Guided Tactical Corridor Policy

The proposed method separates interaction reasoning from dynamic execution by placing a time-varying feasible corridor at their interface. This choice is deliberate. A discrete maneuver label does not specify the lateral space that the planner may use, a target point does not preserve feasibility between updates, and a single learned trajectory couples tactical prediction to one

particular downstream solver. In contrast, the corridor exposes the tactical decision as explicit Frenet inequalities while allowing the common path, speed, and control modules to determine how the maneuver is executed. The method has three coupled elements. First, a Stackelberg–IBR model assigns interaction-aware value to candidate feasible sets. Second, a distributional actor–critic amortizes that game and maps the current duel and track preview to corridor parameters. Third, persistent hybrid logic converts each policy sample into temporally consistent bounds. The following subsections describe these elements in the same order as the online information flow.

3.1. Stackelberg–IBR feasible-set decision

The learned decision is a planner-compatible feasible set rather than a steering or longitudinal command. At tactical step t , the encoder combines ego and opponent states, track preview, the current mode, and the locked pass side. The policy action is projected onto admissible bounds; a deterministic finite-state machine (FSM) and corridor carver then publish

$$\begin{aligned}\tilde{\mathbf{o}}_t &= \mathcal{E}(\mathbf{x}_{e,t}, \mathbf{x}_{o,t}, \mathcal{P}, m_t, \sigma_t^{\text{lock}}), \\ \mathbf{a}_t &= \Pi_{\mathcal{A}}(\pi_{\theta}(\tilde{\mathbf{o}}_t)), \quad (m_{t+1}, \sigma_{t+1}^{\text{lock}}) = \mathcal{F}_m(\mathbf{a}_t, \mathbf{o}_t), \\ \mathcal{G}_t &= (m_{t+1}, \sigma_{t+1}^{\text{lock}}, V_{\text{cap},t}, C_t), \quad C_t = \mathcal{F}_c(\mathcal{B}, \mathbf{x}_{o,t}, \mathbf{a}_t, m_{t+1}),\end{aligned}\quad (23)$$

where $\mathcal{B}(s) = [n_{\text{trk}}^-(s), n_{\text{trk}}^+(s)]$ and $C_t = \{(s, n) : n_t^-(s) \leq n \leq n_t^+(s)\}$. Only π_{θ} is learned; projection, mode persistence, footprint exclusion, and corridor filtering are deterministic and independently inspectable. Equation (23) is therefore an interface contract: the policy may change the side preference, usable width, clearance, and speed cap, but it cannot bypass the track boundary or directly alter an actuator. The contract also makes tactical alternatives comparable before a path is chosen. Two actions can reserve different sides or recovery margins even when their short-horizon centre trajectories appear similar, which is important near an opponent where locally smooth trajectories can imply very different future passing options.

The interaction prior is obtained from a leader–follower game. For vehicle i , a hybrid action contains a maneuver/side choice d_i and continuous aggressiveness/preference variables c_i . Its utility collects progress, relative race position, safety, terminal recoverability, and action regularity in one expression,

$$U_i = w_i^p \Phi_i^p + w_i^r \Phi_i^r - w_i^s \Phi_i^s + w_i^t \Phi_i^t - w_i^c \Phi_i^c. \quad (24)$$

For a fixed discrete profile, the opponent response is approximated by iterative best response (IBR), and the ego selects the worst-case value in a local response neighborhood:

$$\begin{aligned}c_i^{(v+1)} &\in \arg \max_{c_i \in C_i(d_i)} U_i(d, c_e, c_i, c_{-i}^{(v)}), \\ \mathbf{a}_{\text{th}}^*(\mathbf{o}) &\in \arg \max_{(d_e, c_e) \in \mathcal{A}_e^{\text{safe}}(\mathbf{o})} \min_{\|c_{-e} - c_{-e}^{\text{IBR}^*}\| \leq \delta} U_e((d_e, c_e), (d_{-e}, c_{-e})).\end{aligned}\quad (25)$$

This game is evaluated during training to generate structural targets. The online actor amortizes repeated IBR solutions and therefore preserves the interaction semantics without placing an equilibrium solver in the real-time control loop. The construction transfers three pieces of structure rather than treating the

game as an independent expert: $\mathbf{a}_{\text{th}}^*$ supplies an action prior, the robust inner minimum supplies a value target, and the decomposition of U_i determines the roles represented in the closed-loop reward. Thus the game specifies which corridor is strategically useful, whereas learning compresses repeated solutions into a fixed-cost forward pass. This division also creates a direct ablation hook: removing game guidance changes the tactical supervision without changing the policy interface or executor.

Physical admissibility acts on all tactical candidates without defining their strategic value. A short preview of a candidate corridor and speed cap gives tangential and normal acceleration demands $(\hat{a}_{T,t,k}, \hat{a}_{N,t,k})$. With the load-dependent limits of Section 2, the common projection set is

$$\begin{aligned}\hat{\rho}_{t,k}(\mathbf{a}) &= \left| \frac{\hat{a}_{N,t,k}}{a_N^{\text{max}}(\hat{V}_{t,k})} \right|^p + \left| \frac{\hat{a}_{T,t,k}}{a_T^{\text{max}}(\hat{V}_{t,k})} \right|^p, \\ \mathcal{A}_e^{\text{safe}}(\tilde{\mathbf{o}}_t) &= \left\{ \mathbf{a} \in \mathcal{A}_e : \max_k \hat{\rho}_{t,k}(\mathbf{a}) \leq 1 \right\}.\end{aligned}\quad (26)$$

An out-of-set candidate is projected or its V_{cap} is reduced before game or critic evaluation. This step links the learned corridor to the vehicle model in Fig. 2; it does not choose the passing side or replace the interaction utility.

3.2. Game-guided distributional policy

The input is a 51-dimensional vector $\tilde{\mathbf{o}}_t = [\mathbf{o}_t^{\text{duel}}, \boldsymbol{\rho}(s_e), \boldsymbol{\rho}(s_o)]$: $\mathbf{o}_t^{\text{duel}} \in \mathbb{R}^{27}$ contains normalized ego/opponent kinematics, relative progress and lateral gap, closing-time features, FSM memory, and race status; each $\boldsymbol{\rho} \in \mathbb{R}^{12}$ contains curvature and usable-width statistics over a 300-m preview. The actor produces the ten physical parameters

$$\mathbf{a}_t = [L_t^{\text{cap}}, R_t^{\text{cap}}, d_{\text{safe}}, V_{\text{cap}}, g_{\text{chase}}, g_{\text{ot}}, g_{\text{abort}}, b_{\text{lat}}, w_{\text{safe}}, q_{\text{side}}]^{\top}, \quad (27)$$

which respectively control lateral capacity, clearance, speed, hysteresis, far-field bias, recovery width, and pass side. A tanh output is affinely scaled to the intervals in Table A.2 in Appendix A; $\Pi_{\mathcal{A}}$ enforces ordered gap thresholds and a non-degenerate corridor before the action reaches the FSM.

The reward mirrors Eq. (24) but is evaluated from closed-loop events and geometry,

$$\begin{aligned}r_t &= w_{\text{ot}} r_{\text{ot}} + w_{\text{ovb}} r_{\text{ovb}} + w_{\text{col}} r_{\text{col}} + w_{\text{lat}} r_{\text{lat}} \\ &\quad + w_{\text{crush}} r_{\text{crush}} + w_{\text{app}} r_{\text{app}} + w_{\text{sm}} r_{\text{sm}} + w_{\text{gs}} r_{\text{gs}} \\ &\quad + w_{\text{sbs}} r_{\text{sbs}} + w_{\text{wide}} r_{\text{wide}}.\end{aligned}\quad (28)$$

The terms encode pass/overtaken events, contact, lateral feasibility, corridor collapse, approach progress, action and corridor smoothness, side-by-side clearance, and retained usable width. Their reported weights are listed in Table A.2 rather than expanded into routine feature formulae in the main text.

Truncated Quantile Critics (TQC) model the return distribution and remove the largest target atoms before the Bellman backup. This directly addresses the optimistic estimates induced by a small number of high-return passes, whereas the game supplies interaction structure that a scalar environment

reward does not encode explicitly. To keep actor and critic evaluation consistent with the shared physical admissibility check, the raw next action is projected by Eq. (26) before the target critics are evaluated,

$$\begin{aligned}\tilde{\mathbf{a}}_{t+1} &\sim \pi_\theta(\cdot \mid \tilde{\mathbf{o}}_{t+1}), \\ \mathbf{a}_{t+1}^F &= \Pi_{\mathcal{A}_e^{\text{safe}}}(\tilde{\mathbf{a}}_{t+1}), \\ \ell_{t+1} &= \log \pi_\theta(\tilde{\mathbf{a}}_{t+1} \mid \tilde{\mathbf{o}}_{t+1}), \\ \mathcal{Y}_t &= \left\{ r_t + \gamma \left[z'_{(k)}(\tilde{\mathbf{o}}_{t+1}, \mathbf{a}_{t+1}^F) - \alpha \ell_{t+1} \right] \right\}_{k=1}^K, \quad K < 2N_q.\end{aligned}\quad (29)$$

The online actor loss likewise evaluates the critics at the projected action, and the projected mean $\mu_\theta^F = \Pi_{\mathcal{A}_e^{\text{safe}}}(\mu_\theta)$ is used for game-prior alignment. Thus high return cannot be assigned to a speed-corridor combination solely by extrapolating beyond the tire-force support represented in Section 2. Discarding the upper tail limits optimistic backups caused by rare, high-return passing sequences, while retaining a distribution rather than a single mean allows the critic to distinguish corridors with similar expected progress but different adverse-return profiles. This is useful in racing interactions, where a small number of late or geometrically interlocked commitments can dominate the outcome of an otherwise fast policy. The training objective adds two game-derived signals to the standard quantile-Huber critic and entropy-regularized actor losses:

$$\begin{aligned}\mathcal{L} &= J_Q(\varphi) + \lambda_\pi J_\pi(\theta) + \lambda_p \|\mu_\theta^F(\tilde{\mathbf{o}}) - \mathbf{a}_{\text{th}}^*(\tilde{\mathbf{o}})\|_{W_c}^2 \\ &\quad + \lambda_g (\widehat{\mathcal{G}}_\psi(\tilde{\mathbf{o}}, \mathbf{a}) - \mathcal{V}_e(\mathbf{a} \mid \tilde{\mathbf{o}}))^2.\end{aligned}\quad (30)$$

The first added term aligns the actor mean with the IBR action, emphasizing safety-critical components through W_c ; the second anchors an auxiliary head to the robust game value. These terms are used only for training. The deployed actor has the same 51-to-10 interface in every scenario. Network, replay, quantile, and target-update settings are reported in Tables A.1 and A.2 in Appendix A. Consequently, game guidance shapes both the action and value representations during learning, whereas truncation controls return overestimation within the off-policy update. The two mechanisms are complementary: the former supplies interaction structure and the latter controls distributional optimism. At deployment, neither adds an iterative solver or an auxiliary prediction call to the tactical update.

3.3. Persistent intention and corridor synthesis

The FSM converts each continuous policy sample into a temporally persistent intent. Its nominal modes are RACELINE, SHADOW, OVERTAKE, HOLD, and DEFEND. A side may be locked only when opponent-excluded width and relative-speed gates are both satisfied; the lock is retained throughout the duel. The complete online conversion is summarized in Algorithm 1.

At station s_k , the carver first intersects the action-capped

track interval with the mode-dependent opponent exclusion.

$$\begin{aligned}\rho_{n,j,k} &= \frac{1}{2} (L_j |\sin \xi_{j,k}| + W_j |\cos \xi_{j,k}|), \quad j \in \{e, o\}, \\ \rho_{\text{exc},k} &= \rho_{n,e,k} + \rho_{n,o,k} + d_{\text{safe}}, \\ \widehat{n}_{t,k} &= \begin{cases} \max\{n_{\text{cap},k}^-, n_{o,k} + \eta_k \rho_{\text{exc},k}\}, & \sigma_t^{\text{lock}} = +1, \\ n_{\text{cap},k}^-, & \sigma_t^{\text{lock}} = -1, \end{cases} \\ \widehat{n}_{t,k}^+ &= \begin{cases} n_{\text{cap},k}^+, & \sigma_t^{\text{lock}} = +1, \\ \min\{n_{\text{cap},k}^+, n_{o,k} - \eta_k \rho_{\text{exc},k}\}, & \sigma_t^{\text{lock}} = -1. \end{cases}\end{aligned}\quad (31)$$

Here $[n_{\text{cap},k}^-, n_{\text{cap},k}^+]$ denotes the track interval after applying L_t^{cap} and R_t^{cap} , and $(L_j, W_j, \xi_{j,k})$ are the dimensions and heading error of vehicle j . Accordingly, $\rho_{n,e,k}$ and $\rho_{n,o,k}$ are the projected lateral half-widths, while d_{safe} is the policy-specified additional clearance. The factor $\eta_k \in [0, 1]$ combines the existing longitudinal fade and ramp, vanishing outside the opponent influence interval and approaching one as the ego vehicle approaches. Because positive n follows the left normal in Eq. (11), $\sigma_t^{\text{lock}} = +1$ raises the lower bound for a left-side pass, while $\sigma_t^{\text{lock}} = -1$ lowers the upper bound for a right-side pass. Equation (31) describes only the core OVERTAKE contraction; SHADOW, HOLD, and DEFEND retain the biases in Algorithm 1. The existing minimum-width check and exponential moving-average filter are applied afterward, and this carver relation is not a formal safety guarantee. If the raw interval $[\widehat{n}_{t,k}^-, \widehat{n}_{t,k}^+]$ satisfies the minimum-width gate, its bounds are filtered as

$$n_{t,k}^\pm = \beta n_{t-1,k}^\pm + (1 - \beta) \widehat{n}_{t,k}^\pm, \quad \widehat{n}_{t,k}^+ - \widehat{n}_{t,k}^- \geq L_{\min}^{\text{corr}}; \quad (32)$$

otherwise the update is rejected and the last feasible corridor is retained. The minimum-width condition protects the path solver from a geometrically empty request, while the filter suppresses update-to-update boundary motion. These spatial checks complement the FSM's temporal hysteresis: the former preserve an executable interval at each station, and the latter prevents the selected interval from alternating between passing sides during one duel.

This separation is the planner-facing interface of the method: a noisy or late policy update cannot command an actuator directly. An inadmissible sample is projected, a collapsed corridor is rejected, and a missing tactical update holds the last feasible guidance for one control cycle.

4. Corridor-Constrained Closed-Loop Execution

The execution layer is common to the proposed method, every baseline, and every ablation. It is specified here because the tactical corridor is meaningful only if it can be converted into a closed-loop vehicle response, but it is not claimed as a separate contribution. At stations s_k , the published bounds are intersected with the track to obtain $\underline{n}_k = \max(n_{t,k}^-, n_{\text{trk},k}^-)$ and $\bar{n}_k = \min(n_{t,k}^+, n_{\text{trk},k}^+)$. With $\mathbf{z}_k = [n_k, \xi_k]^\top$, sequential iteration j linearizes Eq. (18) as $\widehat{\mathbf{f}}_{s,k}^j = \mathbf{A}_k^j \mathbf{z}_k + \mathbf{B}_k^j \omega_k + \mathbf{b}_k^j$ and imposes the trapezoidal spatial dynamics

$$\mathbf{z}_{k+1} - \mathbf{z}_k = \frac{\Delta s_{\text{sta}}}{2} \left[\widehat{\mathbf{f}}_{s,k}^j(\mathbf{z}_k, \omega_k) + \widehat{\mathbf{f}}_{s,k+1}^j(\mathbf{z}_{k+1}, \omega_{k+1}) \right]. \quad (33)$$

Algorithm 1 Persistent tactical-corridor update

Require: $\mathcal{B}$, ego/opponent states, previous $(m, \sigma^{\text{lock}}, C)$, map preview $\mathcal{P}$

- 1: Encode $\tilde{o}$; sample and project $\mathbf{a} \leftarrow \Pi_{\mathcal{A}}(\pi_{\theta}(\tilde{o}))$
 - 2: Compute relative gap, closing condition, and free width on each side
 - 3: **if** inside overtake gate and selected side is geometrically ready **then**
 - 4: set $m \leftarrow \text{OVERTAKE}$ and lock the selected side
 - 5: **else if** inside chase gate **then**
 - 6: set $m \leftarrow \text{SHADOW}$
 - 7: **else if** abort or post-pass condition is active **then**
 - 8: set $m \leftarrow \text{HOLD}$
 - 9: **else**
 - 10: select RACELINE/DEFEND from front/rear race state
 - 11: **end if**
 - 12: Initialize each station from track bounds capped by $L^{\text{cap}}, R^{\text{cap}}$
 - 13: Exclude the opponent footprint enlarged by d_{safe} ; apply the mode/side bias and V_{cap}
 - 14: Reject corridors narrower than $L_{\text{min}}^{\text{corr}}$; otherwise apply the exponential moving-average filter
 - 15: **return** $\mathcal{G} = (m, \sigma^{\text{lock}}, V_{\text{cap}}, C)$
-

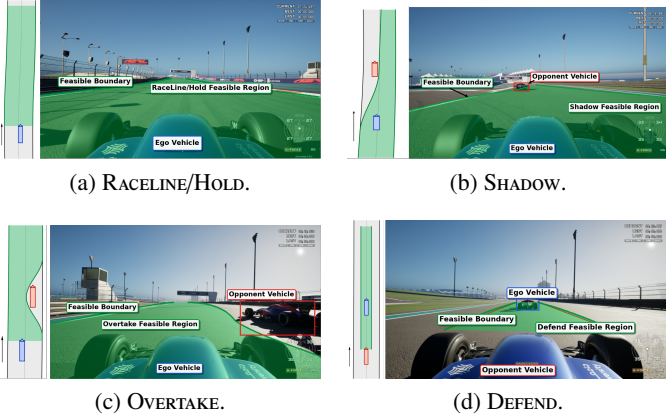


Figure 3: Geometric semantics of the persistent modes. Green regions are the published feasible corridors. SHADOW prepares the locked side, OVERTAKE excludes the opponent footprint on that side, and DEFEND reserves the blocking corridor; the common executor receives only the resulting bounds and speed cap.

The path is the solution of the second-order cone program

$$\begin{aligned}
 \mathbf{y}^{j+1} = \arg \min_{\mathbf{y}} \quad & \sum_{k=0}^{N-1} w_{\omega} \sigma_k^{\omega} + w_{n,T} \epsilon_n + w_{\xi,T} \epsilon_{\xi} \\
 \text{s.t. } \quad & \mathbf{z}_0 = [n_e, \xi_e]^{\top}, \quad \text{Eq. (33)}, \\
 & \underline{n}_k \leq n_k \leq \bar{n}_k, \quad \|[2\omega_k, \sigma_k^{\omega} - 1]^{\top}\|_2 \leq \sigma_k^{\omega} + 1, \\
 & |n_{N-1} - n_T| \leq \epsilon_n, \quad |\xi_{N-1} - \xi_T| \leq \epsilon_{\xi}, \quad \epsilon_n, \epsilon_{\xi} \geq 0,
 \end{aligned} \tag{34}$$

where $\mathbf{y} = \{z_k, \omega_k, \sigma_k^{\omega}, \epsilon_n, \epsilon_{\xi}\}$ and n_T is the raceline or corridor midpoint selected by the persistent mode. The cone is the epigraph of ω_k^2 ; hence the cost regularizes curvature while the terminal slack prevents numerical infeasibility in a short, tightly

carved corridor.

After reconstructing the Cartesian path, let $\kappa_{p,k} \approx \omega_k^*$ and let $a_{T,k}^+(V)$ and $b_{T,k}^-(V) > 0$ denote the available propulsion and braking magnitudes after applying Eqs. (21) and (22). The curvature ceiling and forward-backward sweep are

$$\begin{aligned}
 \bar{V}_{\text{lim},k} &= \begin{cases} v_{x,\text{max}}, & |\kappa_{p,k}| < \varepsilon_{\kappa}, \\ \lim_{r \rightarrow \infty} V_k^{(r)}, & \text{otherwise,} \end{cases} \\
 V_k^{(r+1)} &= \min \left\{ v_{x,\text{max}}, \sqrt{\frac{a_N^{\text{max}}(V_k^{(r)})}{|\kappa_{p,k}|}} \right\}, \quad V_k^{(0)} = v_{x,\text{max}}, \\
 V_{\text{lim},k} &= \min\{\bar{V}_{\text{lim},k}, V_{\text{cap},k}\}, \\
 V_0^f &= \text{clip}_{[V_{\text{min}}, V_{\text{lim},0}]}(V_e), \\
 V_{k+1}^f &= \text{clip}_{[V_{\text{min}}, V_{\text{lim},k+1}]} \sqrt{(V_k^f)^2 + 2\Delta \ell_k a_{T,k}^+(V_k^f)}, \\
 V_{N-1}^b &= \min\{V_{N-1}^f, V_{\text{lim},N-1}\}, \\
 V_k^b &= \text{clip}_{[V_{\text{min}}, V_{\text{lim},k}]} \sqrt{(V_{k+1}^b)^2 + 2\Delta \ell_k b_{T,k+1}^-(V_{k+1}^b)}, \\
 V_k^* &= \min\{V_k^f, V_k^b, V_{\text{lim},k}\}.
 \end{aligned} \tag{35}$$

The curvature ceiling is evaluated by fixed-point iteration; the subsequent passes enforce reachable acceleration and braking on the optimized path. The resulting path-speed sequence is resampled at the controller period T_c . The forward pass is anchored at the measured ego speed after applying the local admissible bounds, whereas the backward pass starts from the reachable terminal speed delivered by the forward pass.

For the MPCC, define $\chi_k = [x_k, y_k, \theta_k]^{\top}$ and $\mathbf{v}_k = [V_k, \delta_k]^{\top}$. The fixed executor employs a reference-conditioned contouring-error model predictive controller, referred to as MPCC hereafter for consistency. Unlike progress-maximizing MPCC formulations that introduce an additional virtual path-progress state, the deployed controller tracks the time-resampled path-speed reference on a fixed prediction grid through contouring-error, heading-error, speed-reference, and steering-reference penalties. Its state and input are $\chi_k = [x_k, y_k, \theta_k]^{\top}$ and $\mathbf{v}_k = [V_k, \delta_k]^{\top}$, respectively. Its identified local prediction model and the principal tracking errors are

$$\begin{aligned}
 \chi_{k+1} &= \chi_k + T_c \begin{bmatrix} V_k \cos \theta_k & V_k \sin \theta_k & \Omega(V_k, \delta_k) \end{bmatrix}^{\top}, \\
 \Omega(V, \delta) &= \frac{\tan(a_{\delta}\delta + b_{\delta})}{L_c} (c_V V + d_V), \\
 e_{c,k} &= (\mathbf{n}_k^r)^{\top} ([x_k - x_k^r, y_k - y_k^r]^{\top}), \quad e_{\psi,k} = \text{wrap}(\theta_k - \theta_k^r).
 \end{aligned} \tag{36}$$

With $\mathbf{e}_{v,k} = \mathbf{v}_k - \mathbf{v}_k^r$, the receding-horizon problem is

$$\begin{aligned}
 \min_{\{\mathbf{v}_k\}} \quad & q_{\Delta\delta} (\delta_0 - \delta_{-1})^2 + \sum_{k=0}^{N_c-1} (q_c e_{c,k+1}^2 + q_{\psi} e_{\psi,k+1}^2 + \mathbf{e}_{v,k}^{\top} \mathbf{R}_v \mathbf{e}_{v,k}) \\
 \text{s.t. Eq. (36),} \quad & V_{\text{min}} \leq V_k \leq v_{x,\text{max}}, \quad |\delta_k| \leq \delta_{\text{max}}, \\
 & |V_k - V_{k-1}| \leq \Delta V_{\text{max}}, \quad |\delta_k - \delta_{k-1}| \leq \dot{\delta}_{\text{max}} T_c.
 \end{aligned} \tag{37}$$

Only the first command is applied. The allocator then maps its longitudinal request and steering angle to throttle, brake pressure, steering-actuator position, and gear.

Algorithm 2 Common corridor-constrained executor

Require: guidance $\mathcal{G}_t$, state $\mathbf{x}_t$, track model, previous path

- 1: Intersect tactical and track bounds; warm-start the previous path
- 2: **for** at most five sequential linearization iterations **do**
- 3: form Eq. (33) and solve SOCP (34); update the linearization path
- 4: **end for**
- 5: reconstruct the Cartesian path and run Eq. (35) under V_{cap}
- 6: resample the path–speed reference and solve MPCC (37); invoke fallback on a failed health gate
- 7: map the first **acceleration/steering speed/steering** command to throttle, brake, steering actuator, and gear; transmit through CAN
- 8: **return** actuator command and feasibility/health status

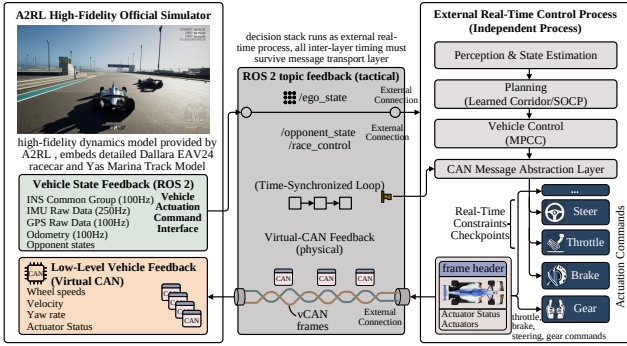


Figure 4: Target-hardware HIL topology. The production algorithm executes on the RGS-8805GC vehicle computer, while the EAV24 plant and track run on a separate server. Vehicle, opponent, and race-control states cross the dual-CAN boundary toward the algorithm. The four actuator channels cross it toward the plant.

Thus the proposed method changes the persistent feasible region and speed cap supplied to a fixed executor, not the executor itself. In Section 5, the corridor/path, mode, and tracking panels test whether the game-layer decision persists and is executed; tire-force-envelope utilization provides the subordinate dynamic-admissibility check.

5. Hardware-in-the-Loop Evaluation

5.1. Target-hardware platform and evaluation protocol

The complete tactical, planning, control, and actuator-allocation stack ran on the Neusys RGS-8805GC industrial computer installed on the A2RL EAV24 platform [19, 17, 20]. The EAV24 plant and Yas Marina environment ran on a separate server (Fig. 4). Ego, opponent, and race-control states entered the target computer through two virtual Controller Area Network (CAN) channels. Throttle percentage, brake pressure, steering-actuator position, and requested gear returned through the same vehicle interface. ROS 2 streams provided instrumentation without replacing this controller-under-test boundary.

The configured plant includes a six-speed mapped powertrain and limited-rate throttle, brake, and steering actuators. It also models suspension, anti-roll, aerodynamic, tire-

relaxation, rolling-resistance, wheel-inertia, thermal, and collision effects. Ego and opponent states are generated at 100 Hz, GNSS/attitude at 100 Hz, and IMU data at 250 Hz. The controller acts at 20 Hz. Camera, lidar, and radar models are configured, but the evaluation uses simulator-provided vehicle states. This choice isolates tactical decision and execution from perception. The setup tests target-hardware timing, transport, solver fallback, actuator mapping, and subsystem-resolved plant response. It does not replace an instrumented track test [18].

Two representative regimes expose complementary tactical demands. Scenario I is a high-speed closure with a short commitment window. Scenario II is a prolonged sharp-corner engagement that demands persistent side selection while both vehicles decelerate and re-accelerate. These runs provide mechanism-level evidence, whereas the matched 1000-lap campaigns provide aggregate evidence. The author-confirmed aggregate schedule was fixed across methods, although its per-lap scenario allocation is unavailable in the current archive. The plant, track, path and speed planners, MPCC, and actuator allocator remained fixed in all matched comparisons.

One closed-loop lap is the aggregate analysis unit. Over-taking success rate (OSR) is the proportion of laps in which longitudinal order changes while lateral clearance is retained within the evaluation horizon. Time to overtake (TTO), per-lap minimum gap, collision count, mean time to collision (TTC), track-boundary violation, and path-feasibility ratio (PFR) provide complementary effectiveness and execution measures.

The principal OSR is reported with a two-sided 95% exact binomial (Clopper–Pearson) interval. This interval treats laps as Bernoulli units and does not model scenario or training-seed clustering. Five independently trained policies provide a separate seed-sensitivity summary as mean $\pm$ standard deviation of seed-level OSRs. Pairwise success-count tests in Tables 2 and 3 use two-sided two-proportion z tests. Holm correction is applied within each four-comparison family. These tests are descriptive because the unavailable per-lap archive prevents hierarchical scenario and seed analysis. The supplied aggregate verifier reproduces all integer counts, confidence intervals, tests, and table entries without reconstructing lap-level observations.

5.2. Closed-loop evidence in two contrasting racing regimes

The two representative runs are deterministic mechanism demonstrations, not independent replicates. Table 1 reports their logged outcomes. Each case follows the same evidence chain. Spatial snapshots first show the tactical opportunity selected by the game-guided policy. Corridor, mode, and cap histories then test whether that decision persists under replanning. Tracking responses test whether the common executor realizes the published feasible set. Relative separation and tire-force utilization provide supporting clearance and physical-admissibility diagnostics. This hierarchy keeps the paper’s main claim on interactive corridor selection rather than on the shared safety and control layers.

Table 1: Closed-loop quantities for the two representative engagements. The overtake interval starts at the first logged OVERTAKE sample and ends at the first later sample with $\Delta s < 0$. RMS errors use every sample in the corresponding case log.

	Scenario I	Scenario II
Tactical stressor	High-speed closure	Sharp-corner side-by-side motion
Interval [s]	1.13	8.63
min Δs [m]	-0.48	-0.84
$V_{\max}$ [m s ⁻¹]	73.10	65.83
RMS e_c [m]	0.02	0.05
RMS e_v [m s ⁻¹]	0.26	1.82

Scenario I: short-window high-speed closure

Scenario I tests whether interaction reasoning reaches the planner before the passing window closes. The ego speed increased from 68.73 to 73.10 m s⁻¹, while the opponent remained near 57.6 m s⁻¹. The FSM first entered SHADOW at 0.74 s and OVERTAKE at 3.19 s. The first later negative Δs occurred at 4.32 s, giving the 1.13-s interval in Table 1.

Figure 5 makes the game-layer decision spatially auditable. At 3.1 s, the policy exposes a continuous corridor and a finite-horizon plan before the ego reaches the opponent. By 4.3 s, the same side is retained as Δs approaches zero. This representation gives the downstream solver a usable region, not only a discrete maneuver label or an instantaneous target point. The two snapshots therefore show the principal advantage of the interface: strategic commitment remains explicit while the executor retains freedom to optimize the path inside it.

Figure 6 connects the tactical decision to closed-loop motion. The path panel samples the latest non-future corridor at each ego station. All 108 samples with an available corridor remained inside it, with a minimum lateral margin of 0.52 m. The speed panel shows sustained closure, whereas acceleration and heading record the commands needed to realize the side change. All 109 acceleration samples satisfy the plotted speed-dependent envelope, with a maximum normalized expression of 0.150. The combined evidence shows that the learned feasible set survives conversion into a dynamically realized reference rather than ending as an unexecuted tactical choice.

Figure 7 tests temporal persistence and execution accuracy. The raw mode record contains one isolated RACELINE sample between two SHADOW segments. The displayed ribbon normalizes this single sample for phase readability, while the source record remains intact. The corridor caps remain between 5.06 and 5.98 m during the maneuver, avoiding a collapsed planning request. The executor achieves 0.020 m lateral RMS and 0.255 m s⁻¹ speed RMS. These responses show that the corridor interface absorbs repeated tactical updates without transferring mode noise directly to the actuators.

Figure 8 provides supporting, rather than organizing, evidence. No recorded sample entered the $|\Delta s| \leq 5$ m and $|\Delta n| \leq 2$ m contact bands simultaneously. When the longitudinal gap was within 5 m, the minimum absolute lateral separation was 5.36 m. The tire panel places every logged acceleration sample inside the fitted envelope. Together, these observations support clearance and model-consistent execution for this run,

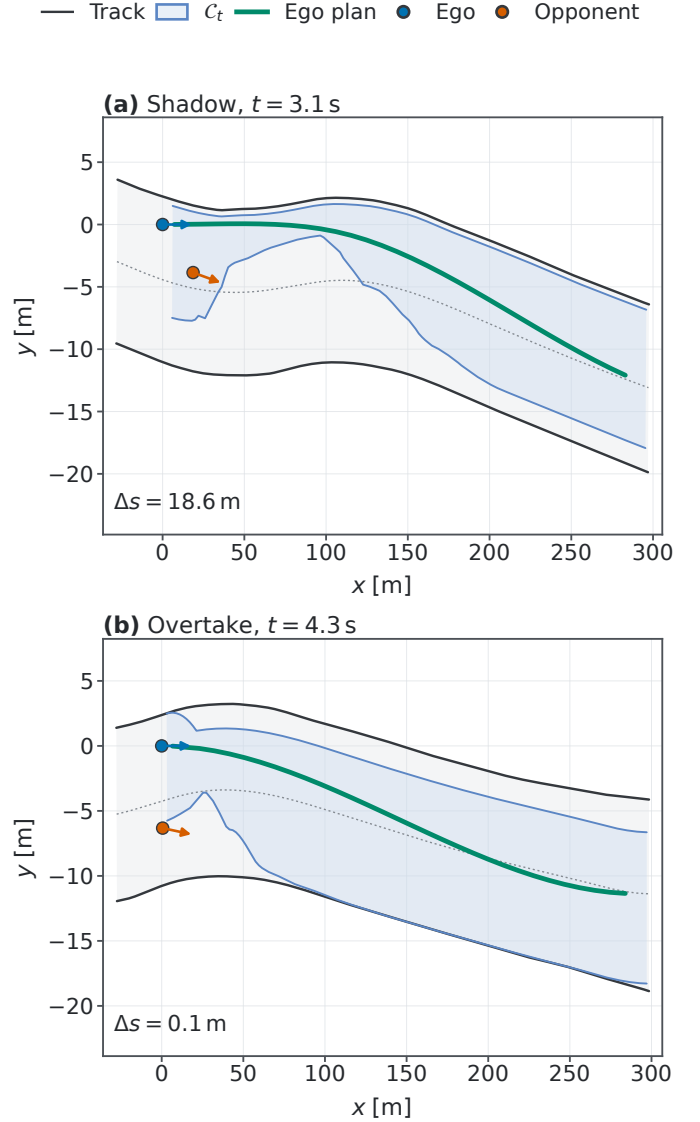


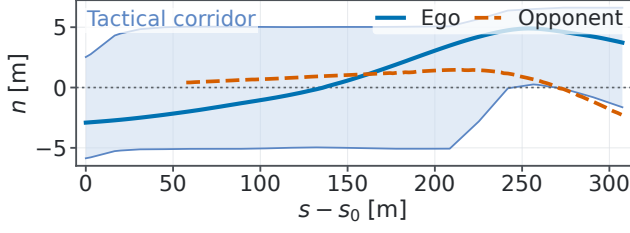
Figure 5: Scenario I interaction geometry. (a) The pre-pass plan is formed inside the game-layer corridor while the vehicles remain separated by $\Delta s = 18.6$ m. (b) The selected side and corridor persist to the order crossover. Blue denotes C_t , green the finite-horizon ego plan, and the arrows vehicle headings.

but they do not constitute a formal safety guarantee.

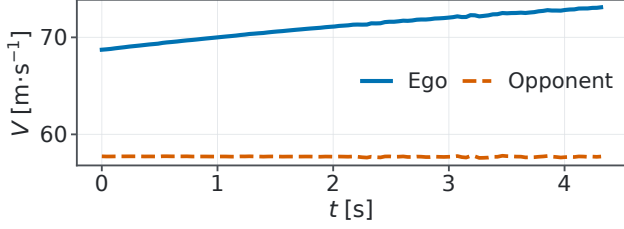
Scenario II: persistent sharp-corner overtake

Scenario II removes the simple advantage of sustained high-speed closure. The ego decelerates from 65.83 to 12.46 m s⁻¹ before accelerating out of the corner, while the opponent follows the same qualitative speed cycle. The FSM enters SHADOW at 0.38 s and OVERTAKE at 2.59 s. It then retains OVERTAKE until the run ends. The first negative Δs sample occurs at 11.22 s, giving an 8.63-s committed interval.

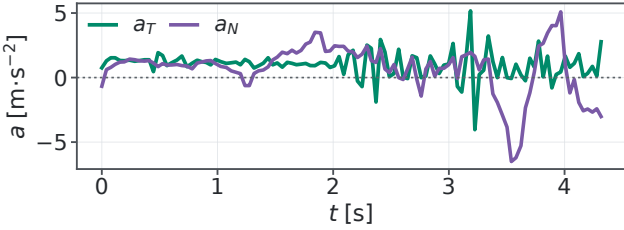
Figure 9 shows why a persistent feasible set matters in this regime. The first snapshot reserves a curved region before the apex, with the opponent 17.8 m ahead. The second preserves a consistent exit-side region as the longitudinal order changes. A transient target point could move across the opponent as curvature and relative speed evolve. In contrast, the published cor-



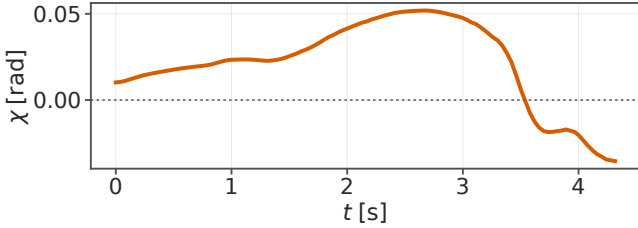
(a) Path and tactical corridor.



(b) Ego and opponent speed.



(c) Tangential and normal acceleration.



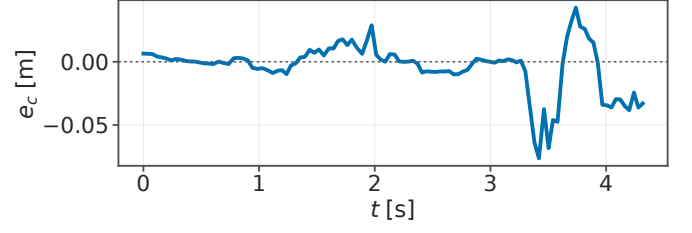
(d) Frenet heading error.

Figure 6: Scenario I closed-loop execution. (a) The ego path remains inside the causal tactical-corridor history. (b) The speed advantage closes the initial gap. (c,d) Acceleration and heading responses show how the common executor realizes the selected side. Panel (c) contains all 109 logged samples.

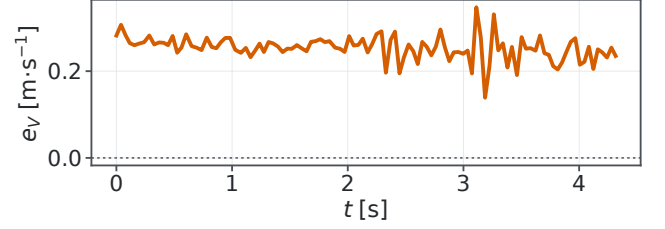
ridor keeps the selected passing opportunity visible across the entire turn and leaves path regularity to the common executor.

Figure 10 shows execution under stronger curvature and speed variation. All 229 samples with an available non-future corridor remain inside the published bounds, with a minimum margin of 0.25 m. The path changes side through the corner while speed falls by more than 50 m s^{-1} and the heading error spans the turn. This continuity is the algorithmic advantage: the game layer reshapes the feasible region while the fixed executor resolves the coupled path and speed response. The acceleration panel is diagnostic and uses the declared envelope display mask; it is not evidence that every plant sample satisfies the reduced envelope.

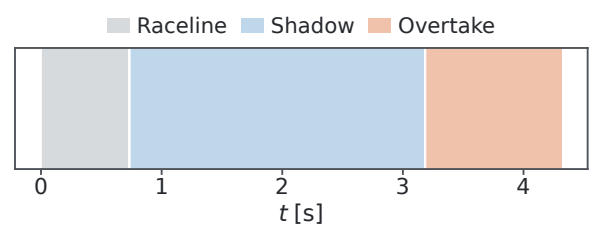
Figure 11 isolates the two mechanisms that sustain the pass. The raw FSM record progresses once from RACELINE to SHADOW



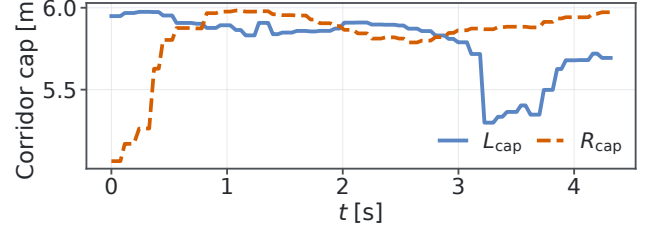
(a) Lateral tracking error.



(b) Speed tracking error.



(c) FSM phase ribbon.

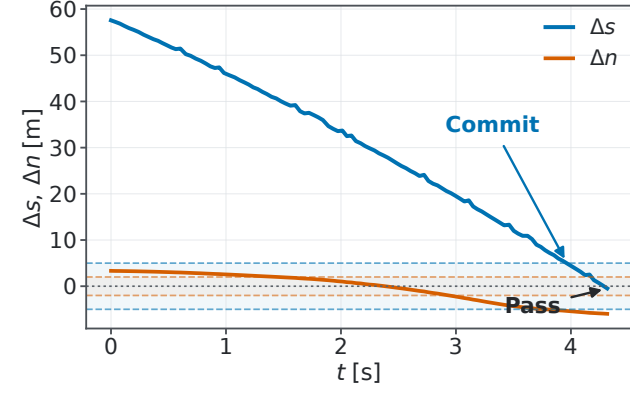


(d) Left and right corridor caps.

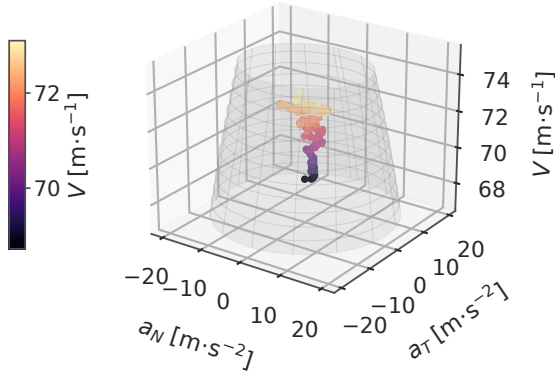
Figure 7: Scenario I tracking and tactical-state responses. (a,b) The common executor tracks the corridor-derived path and speed references. (c) The phase sequence exposes the transition from approach to committed overtake. One of 109 raw mode samples is normalized for display as described in the text. (d) The dimensional corridor caps remain non-degenerate throughout the pass.

and then to OVERTAKE, without display normalization. The left and right caps vary from 3.03–5.99 m and 2.14–6.00 m, respectively. This asymmetric adaptation retains usable width while the track geometry changes. Despite the extended maneuver, lateral RMS remains 0.051 m, with a 0.147-m maximum absolute error. Speed RMS is 1.82 m s^{-1} , with a 3.50 m s^{-1} maximum. Thus the adaptive and persistent corridor reaches the vehicle as a trackable reference rather than repeatedly restarting the planner.

Figure 12 separates clearance evidence from the limits of the reduced physical model. No sample enters both rectangular contact bands; when $|\Delta s| \leq 5 \text{ m}$, the minimum $|\Delta n|$ is 5.26 m. This supports clearance during the order change. The tire panel shows only the 193 samples that satisfy $\rho \leq 1$ under the fit-



(a) Relative longitudinal and lateral separation.



(b) Tire-force envelope utilization.

Figure 8: Scenario I supporting clearance and physical-admissibility checks. (a) The shaded threshold bands show the rectangular contact test; Commit and Pass mark the final +5-m crossing and the first negative Δs sample. (b) All 109 logged acceleration samples are shown relative to the speed-dependent tire-force envelope.

ted envelope. The remaining 37 of 230 logged samples are excluded by the plotting rule, and the maximum raw expression is 2.284. Consequently, the panel describes the in-envelope operating distribution but cannot certify universal closed-loop envelope satisfaction. This discrepancy bounds the physical-admissibility claim without altering the evidence for persistent tactical-corridor execution.

Across the repeated evaluation, the full method completed 947 of 1000 laps (OSR 94.7%, 95% exact binomial confidence interval 93.1–96.0%), with no observed collision or track-boundary violation. Across five independently trained policies, seed-level OSR was $94.3\% \pm 0.7\%$ (mean $\pm$ standard deviation). The following matched baselines and ablations test whether these aggregate outcomes are attributable to the tactical representation rather than the shared executor.

5.3. Matched tactical baselines

Only the tactical-guidance module changes across methods. All methods use the same plant, race-control interface, corridor projection, path and speed planners, MPCC, and actuator allocator. *Pure Raceline* contains no opponent-aware tactical decision. *Non-Interactive MPC* predicts the opponent in-

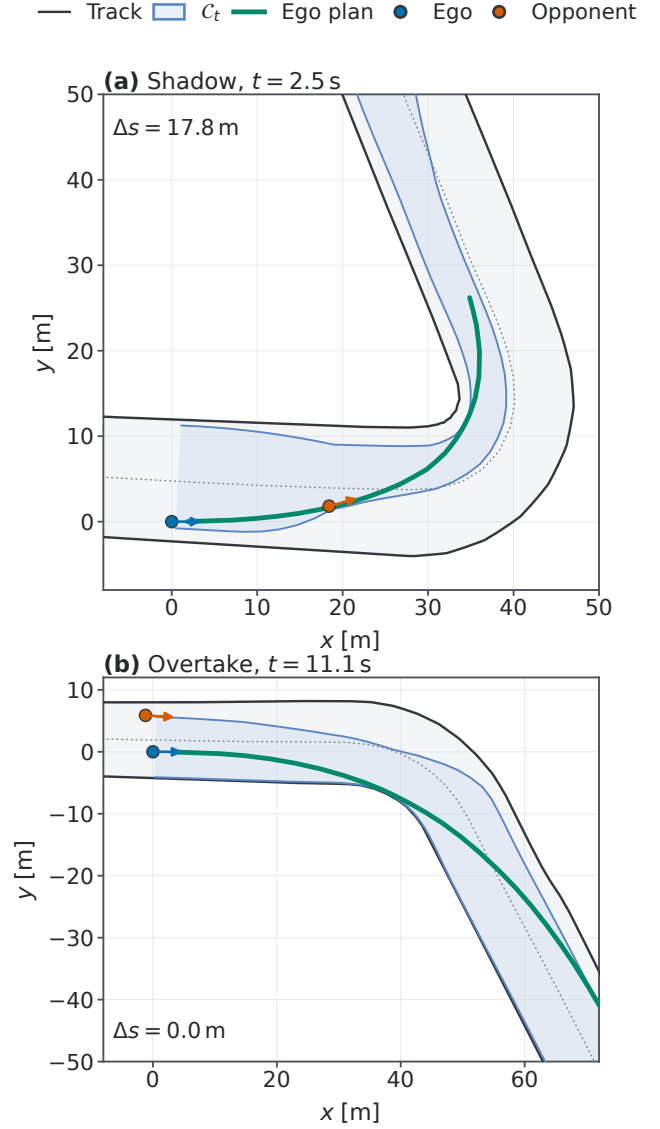
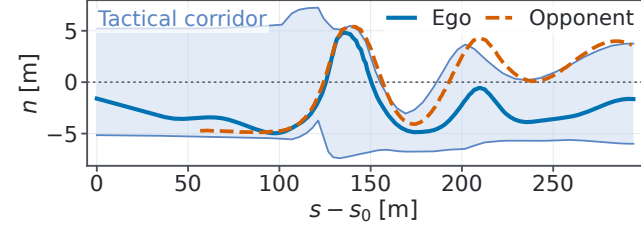


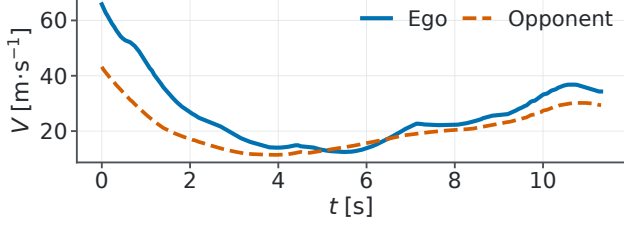
Figure 9: Scenario II interaction geometry. (a) The corridor and ego plan prepare the sharp-corner pass with $\Delta s = 17.8$ m. (b) The same tactical side remains available at the corner exit as longitudinal order changes.

dependently of the ego plan and imposes obstacle constraints on the result [2, 4]. *Rule-based GT* uses a fixed-threshold state machine ($g_{\text{chase}} = 20$ m, $g_{\text{ot}} = 8$ m, and $g_{\text{abort}} = 35$ m). *iLQGames* computes coupled local responses with iterative linear-quadratic game approximations [8]. Outputs from all baselines are projected onto the same side, speed-cap, and corridor interface so that the reported differences concern tactical reasoning rather than low-level execution.

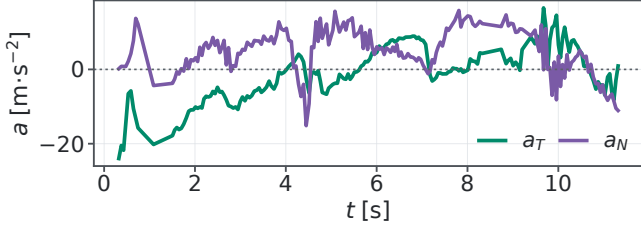
Table 2 makes the primary effect explicit through the OSR and OSR-deficit rows. The proposed tactical corridor policy completed 94.7% of the laps, 31.2 percentage points more than the strongest baseline, *iLQGames*. It also recorded no collision and retained a feasible path in every planning cycle. *iLQGames* confirms the value of interactive reasoning relative to the non-interactive optimizer, but its 63.5% OSR and 14 contacts show that an online game trajectory alone did not provide the per-



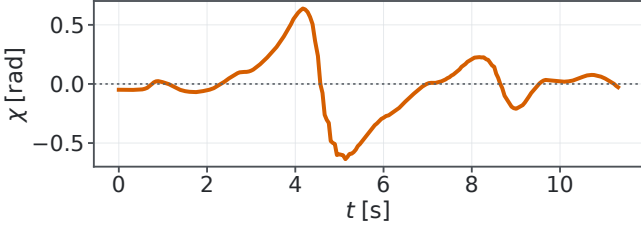
(a) Path and tactical corridor.



(b) Ego and opponent speed.



(c) Tangential and normal acceleration.

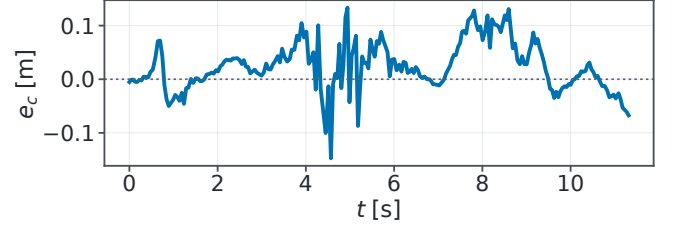


(d) Frenet heading error.

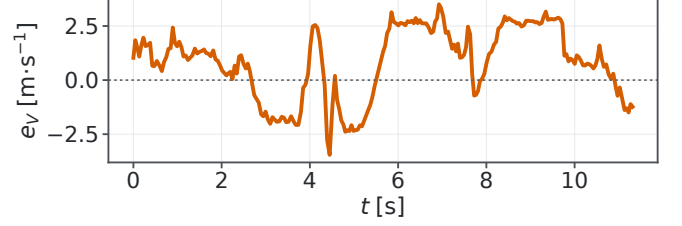
Figure 10: Scenario II closed-loop execution. (a) The realized path stays inside the causal corridor history through the corner. (b,d) Speed and heading expose the braking, rotation, and exit phases. (c) Acceleration displays the 193 of 230 samples satisfying the plotted envelope rule; 37 omitted samples are connected across in recorded time, and the immutable log retains all samples.

sistence of the learned corridor interface. Conversely, Non-Interactive MPC maintained the second largest mean minimum gap and only seven contacts but had the lowest OSR. The combination is evidence of conservative tactical freezing rather than poor low-level tracking. The rule-based method produced 48 contacts, showing that one threshold set did not transfer across the matched curvature and relative- speed regimes. These contrasts isolate the intended gap: interaction reasoning must be both opponent-aware and expressed as a persistent, adaptable feasible region, rather than as an isolated game trajectory, obstacle constraint, or fixed maneuver threshold.

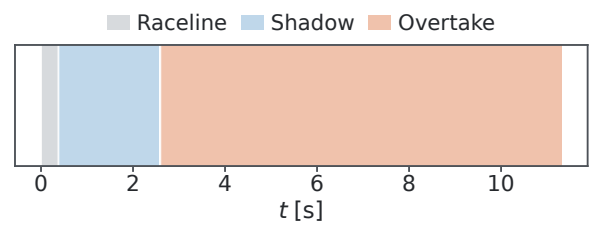
For transparency, two-sided two-proportion z tests were recomputed from the machine-readable integer success counts and adjusted across the four baseline comparisons by the Holm



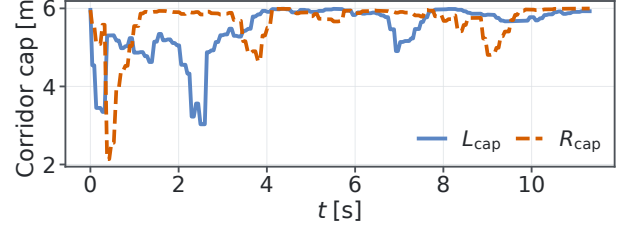
(a) Lateral tracking error.



(b) Speed tracking error.



(c) FSM phase ribbon.



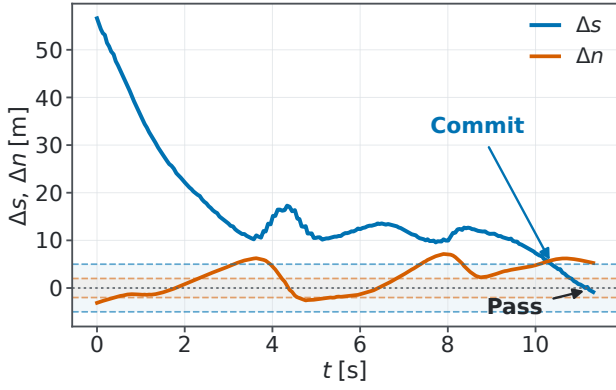
(d) Left and right corridor caps.

Figure 11: Scenario II tracking and tactical-state responses. (a,b) Tracking remains bounded through braking, cornering, and acceleration. (c) The raw mode sequence shows uninterrupted OVERTAKE commitment after 2.59 s. (d) The changing left and right caps expose the learned corridor adaptation that is hidden by a discrete maneuver label.

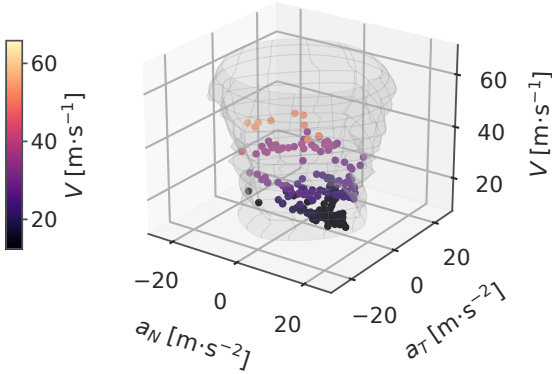
procedure. All count-level differences remained below $p = 10^{-6}$. These tests do not model scenario or training-seed clustering and are therefore treated as descriptive support; the effect sizes and event counts in Table 2 carry the primary interpretation.

5.4. Ablation of the tactical representation

Four ablations isolate the proposed tactical components while retaining the same execution layer. *Without game guidance* removes both the IBR action prior and the auxiliary robust-value target. *Without FSM* bypasses hysteretic mode and side locking, allowing the policy to reshape the corridor on every update. *Without learned corridor* replaces the continuous width, clearance, lateral-bias, and speed-cap outputs by



(a) Relative longitudinal and lateral separation.



(b) In-envelope tire-force utilization.

Figure 12: Scenario II clearance and tire-force diagnostics. (a) Relative separation remains outside simultaneous contact thresholds during the pass. (b) The surface and points show the 193 of 230 logged samples satisfying the displayed $\rho \leq 1$ envelope. The 37 out-of-envelope samples remain in the source log and are not plotted; this panel is not a hard safety certificate.

Table 2: Matched comparison over 1000 closed-loop HIL laps per method. Arrows indicate the preferred direction.

Metric	Prop.	PR	NI-MPC	RB-GT	iLQG
OSR [%] ↑	94.7	38.2	28.5	51.6	63.5
OSR def. [pp] ↓	0.0	56.5	66.2	43.1	31.2
TTO [s] ↓	4.82	7.25	11.23	9.37	7.83
Min. gap [m] ↑	5.83	1.42	3.85	2.14	2.31
Coll. [count] ↓	0	21	7	48	14
Mean TTC [s] ↑	8.52	3.21	6.92	5.64	5.73
Track viol. [%] ↓	0.0	0.0	0.0	0.0	0.0
PFR ↑	1.000	0.988	0.998	0.987	0.995

Notes: Prop., proposed method; PR, Pure Raceline; NI-MPC, Non-Interactive MPC; RB-GT, Rule-based GT; iLQG, iLQGames. OSR def. is the percentage-point deficit from Prop. Min. gap is the mean per-lap minimum inter-vehicle distance. PFR is the fraction of planning cycles with a feasible SOCP solution. Bold denotes the best observed value; equal track violation values are jointly bold.

fixed nominal values. The final variant replaces TQC with Soft Actor-Critic (SAC) while preserving the network, replay, reward, and other training settings.

Table 3 shows that continuous corridor adaptation produced the largest measured effect. Fixing the corridor reduced OSR by 18.6 percentage points, increased contacts from 0 to 31, and gave the lowest PFR. Removing the FSM reduced OSR by 12.4

Table 3: Component ablations over 1000 closed-loop HIL laps per variant. Each variant removes or replaces one tactical component. Arrows indicate the preferred direction.

Metric	Full	No-GG	No-FSM	Fixed-C	SAC
OSR [%] ↑	94.7	85.4	82.3	76.1	88.5
OSR loss [pp] ↓	0.0	9.3	12.4	18.6	6.2
TTO [s] ↓	4.82	6.38	6.71	8.94	5.53
Coll. [count] ↓	0	9	12	31	5
Min. gap [m] ↑	5.83	2.86	2.17	1.05	3.94
Mean TTC [s] ↑	8.52	4.93	4.31	2.86	6.17
PFR ↑	1.000	0.983	0.974	0.961	0.992

Notes: Full, full method; No-GG, without game guidance; No-FSM, without the finite-state machine; Fixed-C, fixed corridor. OSR loss is the percentage-point difference from Full. PFR is the fraction of planning cycles with a feasible SOCP solution. Bold denotes the best observed value.

points and introduced 12 contacts, consistent with corridor revisions that are not held long enough for the receding-horizon executor. Removing game guidance reduced OSR by 9.3 points and increased TTO by 1.56 s. Thus the environment reward alone did not reproduce the timing and clearance obtained with the IBR-consistent prior and auxiliary target. Replacing TQC with SAC produced the smallest OSR reduction, 6.2 points, but introduced five contacts. This outcome is consistent with, but does not by itself prove, the proposed role of truncated return quantiles in avoiding low-probability adverse corridors.

Two-sided count-level proportion tests with Holm adjustment across the four ablations gave adjusted $p < 10^{-6}$ for OSR in every case. As in the baseline comparison, these values are descriptive because only aggregate counts are included in the present statistical record. The ablation effect sizes, collision counts, and PFR changes jointly support the conclusion that the tactical corridor representation is the main source of the measured gain. The ordering of the drops maps onto the proposed mechanism: adaptive corridor shaping creates the opportunity, the FSM preserves it long enough to execute, game guidance improves its timing and clearance, and TQC controls optimistic distributional updates.

5.5. Failure modes and validity boundary

The comparisons expose the pain points targeted by the method. Non-Interactive MPC retained clearance but missed transient passing windows. The fixed corridor could not reshape the usable track region, and non-persistent guidance revised intent faster than the executor could absorb. These failures reduce OSR and PFR even though the plant, tire-force limits, planner, and controller are shared. The full method's 53 non-successes were non-completions within the evaluation horizon. The available aggregate record does not support a finer causal partition. Zero observed collisions therefore means zero events in these 1000 laps, not a formal safety guarantee.

The two case records also expose narrower evidence boundaries. One Scenario I mode sample is normalized only in the display ribbon. The raw value remains in the archived log. In Scenario II, 37 measured acceleration samples exceed the fitted reduced-order envelope and are omitted by the supplied plotting mask. This prevents interpreting the tire panel as universal plant-level constraint satisfaction. It does not change the logged

corridor containment, mode persistence, tracking, or separation results. Future work should reconcile the preview envelope with measured high-fidelity plant accelerations and should show every point or a formally justified residual diagnostic.

The present evidence is bounded to two-vehicle, simulator-state-informed HIL. It does not establish robustness to missed detections, occlusion, multiple opponents, changing grip, tire degradation, damage, or opponents outside the training behavior distribution. Moreover, the archived aggregates do not retain the per-lap scenario and seed map needed for clustered uncertainty or paired lap-level effect intervals. These are explicit targets for the next campaign and for controlled EAV24 track validation.

6. Discussion

The results support a specific architectural claim rather than a general claim that learning should replace model-based racing control. The proposed policy performs the part of the problem for which interaction data are most valuable: selecting when to commit, which side to preserve, and how much space and speed to expose to the executor. The deterministic corridor then acts as a contract. It makes the learned decision visible to the path solver, allows infeasible intervals to be rejected or held, and prevents policy noise from appearing directly at the actuators. This interpretation is consistent with the matched comparison, where every method used the same planner and controller but produced markedly different tactical outcomes.

The tire-force model is necessary to keep candidate corridors and speed caps within the identified vehicle capability, and the envelope figures verify that the representative maneuvers remain executable. It is not the source of the comparative gain: all matched methods share the same physical-feasibility projection, speed planner, and controller. The causal evidence instead points to how the passing opportunity is represented, guided, and preserved across updates.

The ablations refine this interpretation. Replacing the adaptive corridor by a fixed one caused the largest loss, so the measured benefit cannot be explained by discrete behavior selection alone. Removing the FSM shows that a geometrically meaningful action still needs temporal persistence. Removing the game-derived targets shows that reward optimization alone did not recover the same timing and clearance, while the smaller TQC–SAC difference indicates that distributional pessimism is complementary rather than the sole source of performance. Together, these observations distinguish the contribution from both direct game solvers [6, 7, 8] and end-to-end racing policies [12, 13, 14].

The HIL setup addresses the practical emphasis of the target application. The algorithm executes on the intended vehicle computer, observes a communication boundary, and drives subsystem-resolved vehicle and actuator dynamics through the EAV24 CAN command set. This is stronger evidence of deployability than an offline trajectory replay because timing, solver fallback, and command allocation remain in the loop. It is nevertheless not a substitute for track experiments. In particular, the present evidence does not quantify a validated sim-

Table A.1: Principal algorithm and closed-loop evaluation settings.

Component	Parameter	Value
Vehicle model	$m; I_f, I_r; T_d$	742 kg; 1.75, 1.35 m; 0.05 s
Tire curves	$F_{z,0}; (C_{\alpha}, C_{\gamma}); (E_{\alpha}, E_{\gamma})$	2079.72 N; (1.6144, 1.5998); (0, 0)
Tire curves	$(\mu_s, \mu_r); \text{load slopes; peak slips}$	(2.2345, 1.9745); (−0.2555, −0.4787); (0.1296, 0.140)
Tire-force envelope	$p; v_{s,\max}$	0.75; 90 m s ^{−1}
Policy	input/output dimensions	51/10
TQC	critics; N_g ; retained atoms K ; γ	2; 25; 46; 0.99
TQC feasibility	actor/target action support	$\max_k \hat{p}_{t,k} \leq 1$
Corridor	preview; stations; spacing	300 m; 150; 2 m
Safeguards	$L_{\min}^{\text{corr}}$; filter β	3 m; 0.2
Path solver	maximum linearization iterations	5
MPCC	steps; sample time; horizon	15; 0.05 s; 0.75 s

to-track error, worst-case computation time, perception uncertainty, or the effects of changing tire and surface conditions.

The immediate research priorities follow from these limits. First, the tactical observation should be driven by a perception-and-prediction stack so that corridor uncertainty expands with opponent-state uncertainty. Second, the game and carver should be generalized from one opponent to several simultaneous response sets. Third, the aggregate evaluation should retain per-lap scenario, training-seed, deadline, and failure-mode identifiers to support hierarchical uncertainty analysis. Finally, controlled EAV24 track tests should compare HIL and measured corridor execution under matched initial conditions. These steps test the transfer boundary without changing the core feasible-set interface.

7. Conclusion

This paper addressed the gap between strategically meaningful overtaking decisions and planner-executable closed-loop behavior. It introduced a game-guided policy that learns a persistent tactical feasible set rather than a low-level control action: Stackelberg–IBR targets amortize repeated game reasoning, TQC limits optimistic distributional updates, and a hysteretic state machine with deterministic carving turns the result into a time-varying Frenet corridor for a common execution layer. On the target RGS-8805GC computer in a dual-CAN EAV24 HIL loop, the method achieved 94.7% overtaking success over 1000 laps, with no observed collision or track violation. Matched baselines and ablations show that adaptive corridor shaping, game-derived guidance, and persistent side commitment each contribute to this result. The evidence supports tactical corridor learning as a practical interface between interaction reasoning and constrained vehicle execution; the tire-force model provides supporting physical admissibility rather than the organizing contribution. This evidence does not establish a formal safety guarantee or robustness beyond two-vehicle, state-informed HIL engagements; perception-aware multi-opponent and on-track tests remain necessary.

Appendix A. Algorithm and Evaluation Parameters

This appendix collects the numerical settings referenced by the vehicle model, tactical policy, corridor generator, path planner, and controller. The main text retains only the dimensions and values needed to explain the algorithmic logic and experimental outcomes.

Table A.2: Tactical-policy bounds, reward weights, and training settings. Symbols follow Eqs. (27) and (28).

Group	Setting
Corridor and speed	$L_t^{\text{cap}}, R_t^{\text{cap}} \in [1.5, 6.0] \text{ m}; d_{\text{safe}} \in [0.2, 0.8] \text{ m}; V_{\text{cap}} \in [15, 90] \text{ m s}^{-1}$
Gap thresholds	$g_{\text{chase}} \in [18, 24] \text{ m}; g_{\text{ot}} \in [16, 24] \text{ m}; g_{\text{abort}} \in [28, 45] \text{ m}$
Bias, recovery, side	$b_{\text{lat}} \in [-1.5, 1.5] \text{ m}; w_{\text{safe}} \in [1, 5] \text{ m}; q_{\text{side}} \in [-1, 1]$
Reward weights	$(W_{\text{ot}}, W_{\text{ovb}}, W_{\text{col}}, W_{\text{lat}}, W_{\text{crush}}, W_{\text{app}}, W_{\text{sm}}, W_{\text{gs}}, W_{\text{abs}}, W_{\text{wide}}) = (30, -20, -100, -2, -5, 3, -0.1, -0.5, 0.5, 0.25)$
Optimization	learning rate 3×10^{-4} ; replay capacity 10^5 ; batch size 256; target update 5×10^{-3}
Networks	feature encoder $51 \rightarrow 256 \rightarrow 256$; actor and critic hidden layers [256, 256]
Software	Python 3.10.0; SB3 2.8.0; PyTorch 2.4.0+cu124; Gymnasium 1.2.3

Acknowledgements

The authors acknowledge the Abu Dhabi Autonomous Racing League for providing the EAV24 simulation and vehicle-interface environment.

References

- [1] J. Betz, A. Wischniewski, A. Heilmeyer, F. Nobis, L. Hermansdorfer, T. Stahl, T. Herrmann, and M. Lienkamp, “A software architecture for the dynamic path planning of an autonomous racecar at the limits of handling,” in *Proc. IEEE Intelligent Vehicles Symposium (IV)*, 2019, pp. 1–8.
- [2] J. Kabzan, M. de la Iglesia Valls, V. J. F. Reijgwart *et al.*, “AMZ Driverless: The full autonomous racing system,” *Journal of Field Robotics*, vol. 37, no. 7, pp. 1267–1294, 2020.
- [3] J. Betz, H. Zheng, A. Liniger *et al.*, “Autonomous vehicles on the edge: A survey on autonomous vehicle racing,” *IEEE Open Journal of Intelligent Transportation Systems*, vol. 3, pp. 458–488, 2022.
- [4] J. Betz, T. Betz, F. Fent *et al.*, “TUM autonomous motorsport: An autonomous racing software for the Indy Autonomous Challenge,” *Journal of Field Robotics*, vol. 40, no. 4, pp. 783–809, 2023.
- [5] C. Jung, A. Finazzi, H. Seong *et al.*, “An autonomous system for head-to-head race: Design, implementation and analysis,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 10, pp. 10470–10486, 2023.
- [6] A. Liniger and J. Lygeros, “A noncooperative game approach to autonomous racing,” *IEEE Transactions on Control Systems Technology*, vol. 28, no. 3, pp. 884–897, 2020.
- [7] M. Wang, Z. Wang, J. Talbot, J. C. Gerdes, and M. Schwager, “Game-theoretic planning for self-driving cars in multivehicle competitive scenarios,” *IEEE Transactions on Robotics*, vol. 37, no. 4, pp. 1313–1325, 2021.
- [8] D. Fridovich-Keil, E. Ratner, L. Peters, A. D. Dragan, and C. J. Tomlin, “Efficient iterative linear-quadratic approximations for nonlinear multi-player general-sum differential games,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 1475–1481.
- [9] F. Laine, D. Fridovich-Keil, C.-Y. Chiu, and C. J. Tomlin, “Multi-hypothesis interactions in game-theoretic motion planning,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 8016–8023.
- [10] Y. Jia, M. Bhatt, and N. Mehr, “RAPID: Autonomous multi-agent racing using constrained potential dynamic games,” in *Proc. European Control Conference (ECC)*, 2023, pp. 1–8.
- [11] D. Kalaria, C. Maheshwari, and S. Sastry, “ α -RACER: Real-time algorithm for game-theoretic motion planning and control in autonomous racing using α -potential function,” in *Proc. Learning for Dynamics and Control Conference*, PMLR, vol. 283, pp. 1–18, 2025.
- [12] P. R. Wurman, S. Barrett, K. Kawamoto *et al.*, “Outracing champion Gran Turismo drivers with deep reinforcement learning,” *Nature*, vol. 602, no. 7896, pp. 223–228, 2022.
- [13] F. Fuchs, Y. Song, E. Kaufmann, D. Scaramuzza, and P. Dür, “Super-human performance in Gran Turismo Sport using deep reinforcement learning,” *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4257–4264, 2021.

- [14] Y. Song, H. Lin, E. Kaufmann, P. Dür, and D. Scaramuzza, “Autonomous overtaking in Gran Turismo Sport using curriculum reinforcement learning,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 9403–9409.
- [15] H. Lin, Y. Song, and D. Scaramuzza, “Reaching the limit in autonomous racing: Optimal control versus reinforcement learning,” *Science Robotics*, vol. 8, no. 82, art. eadg1462, 2023.
- [16] R. S. Thakkar, A. S. Samyal, D. Fridovich-Keil, Z. Xu, and U. Topcu, “Hierarchical control for head-to-head autonomous racing,” *Field Robotics*, vol. 4, no. 1, pp. 46–69, 2024, doi: 10.55417/FR.2024002.
- [17] S. Hoffmann, S. Sagmeister, T. Betz *et al.*, “Head-to-head autonomous racing at the limits of handling in the A2RL challenge,” arXiv preprint arXiv:2602.08571, 2026.
- [18] S. Sagmeister, P. Kounatidis, S. Goblirsch, and M. Lienkamp, “Analyzing the impact of simulation fidelity on the evaluation of autonomous driving motion control,” in *Proc. IEEE Intelligent Vehicles Symposium (IV)*, 2024, pp. 230–237, doi: 10.1109/IV55156.2024.10588858.
- [19] Abu Dhabi Autonomous Racing League (A2RL), “Autonomous car race 2024,” 2024. [Online]. Available: <https://a2rl.io/autonomous-car-race/car-race-2024> (accessed 18 August 2026).
- [20] Neousys Technology Inc., “RGS-8805GC AMD EPYC 7003 series rugged HPC server: Product datasheet,” 2023. [Online]. Available: <https://www.neousys-tech.com/product/product-lines/rugged-industrial-hpc-server/rgs-8805gc>.